

Dify 可观测

大模型应用开发和全链路可观测实践

郑立

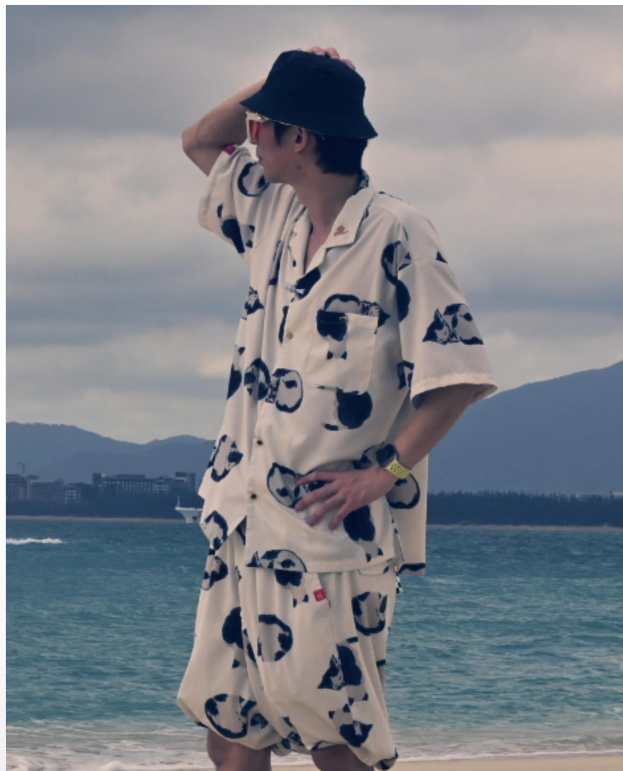
资深开发者关系

2025/11/15

施子腾

阿里云可观测工程师





郑立（布拿拿）

Dify Coder, Dev&Rel, Issue & PR terminator.

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- 01 Dify 简介
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Global Growth & Enterprise Readiness

2.5

Years

5

Physical Offices

10 x

Revenue Growth

70 +

Talents

200 +

Global Enterprise and MNC Clients

NORTH AMERICA

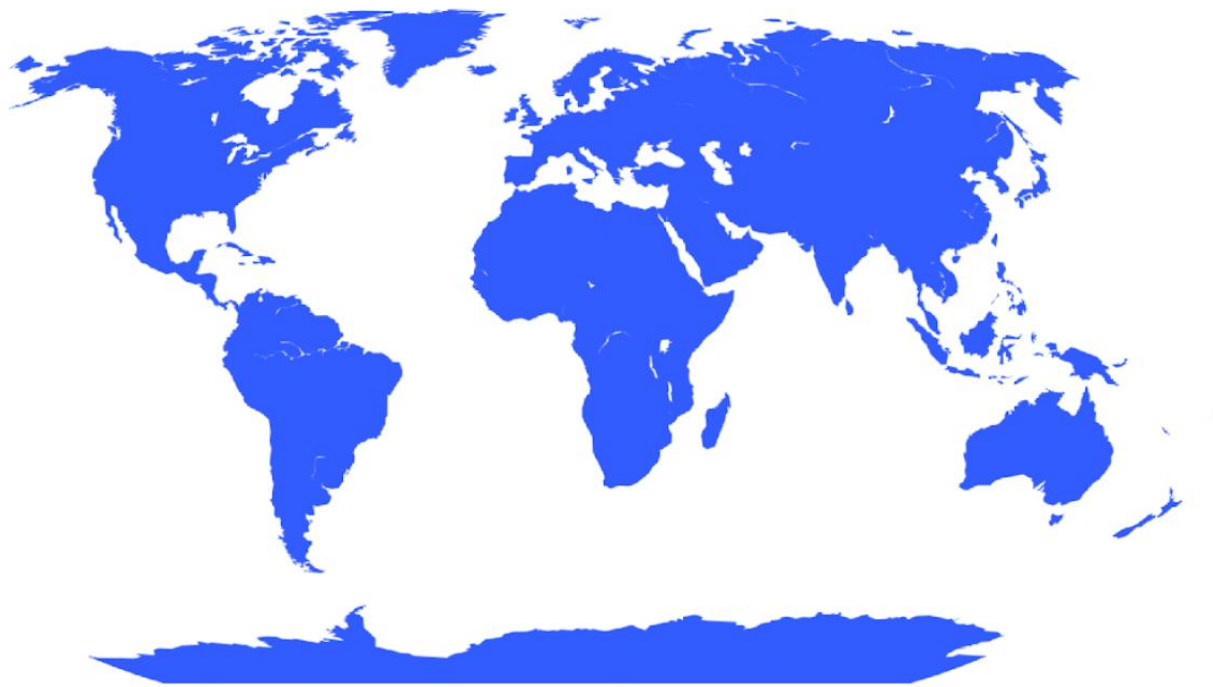
- SUNNYVALE

EUROPE

- PARIS

EAST ASIA

- TOKYO
- SHANGHAI
- SUZHOU



As the **47th** most popular open source project of all time, we have worked with many companies in **putting AI to real work.**

POPULARITY

110,000+

GitHub Stars

OPEN-SOURCE CONTRIBUTION

800+

Contributors

LOCAL DEPLOYMENT

5,500,000+

Docker Pulls

BUILD APPS

1,000,000+

Apps Powered by Dify

Trusted and Use by
Leading Enterprises Across Industries Worldwide

VOLVO

Panasonic

Infineon

Yum!

Mondelēz
International

MAERSK

NOVARTIS

P&G

價格.com

Lilly

NTTEAS

ETS

GovTech 東京

Cartier

T D S E

dip
日本電子計算株式会社

ANKER

Deloitte

ThermoFisher
SCIENTIFIC

VW

BAYER

ADVANTECH

McDonald's

RICOH

CEQUENS

MUFG

Daiichi-Sankyo

AIA

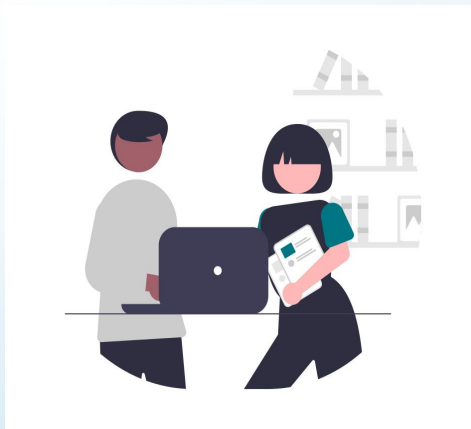
Identify Use Cases

Dify 多功能特性涵盖了众多业务领域，能够推动创新并提升效率，且这种效果在最初的部署之后仍能持续显现。

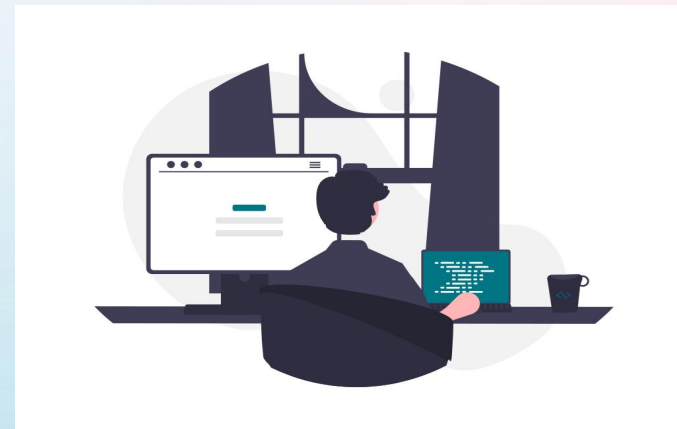
Dify



Industry Specific Use Cases



Departmental Use Cases



Personal Use Cases

Architectural Blueprint

600

QPS Scalability Leadership

Handled with 20 API pods and 16 Sandbox pods, supporting enterprise-wide AI deployment at scale.

40

Resource Mastery (Pods)

Balanced CPU and RAM across pods, enabling seamless scaling and efficient resource utilization.

3s

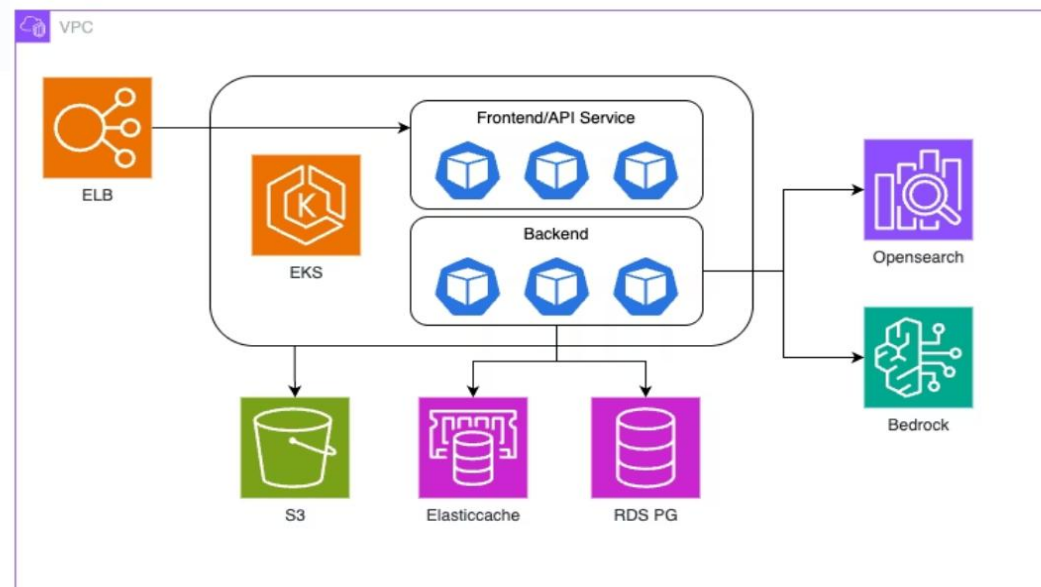
P99 Latency Excellence

Reduced P99 latency from 90s under 500 rps, optimizing user experience with Gunicorn Gevent and 256 ACU.

99.9%

Resilience Strength (Uptime)

Achieved with HA geo-redundant replicas and database backups, ensuring business continuity (5-minute RTO).



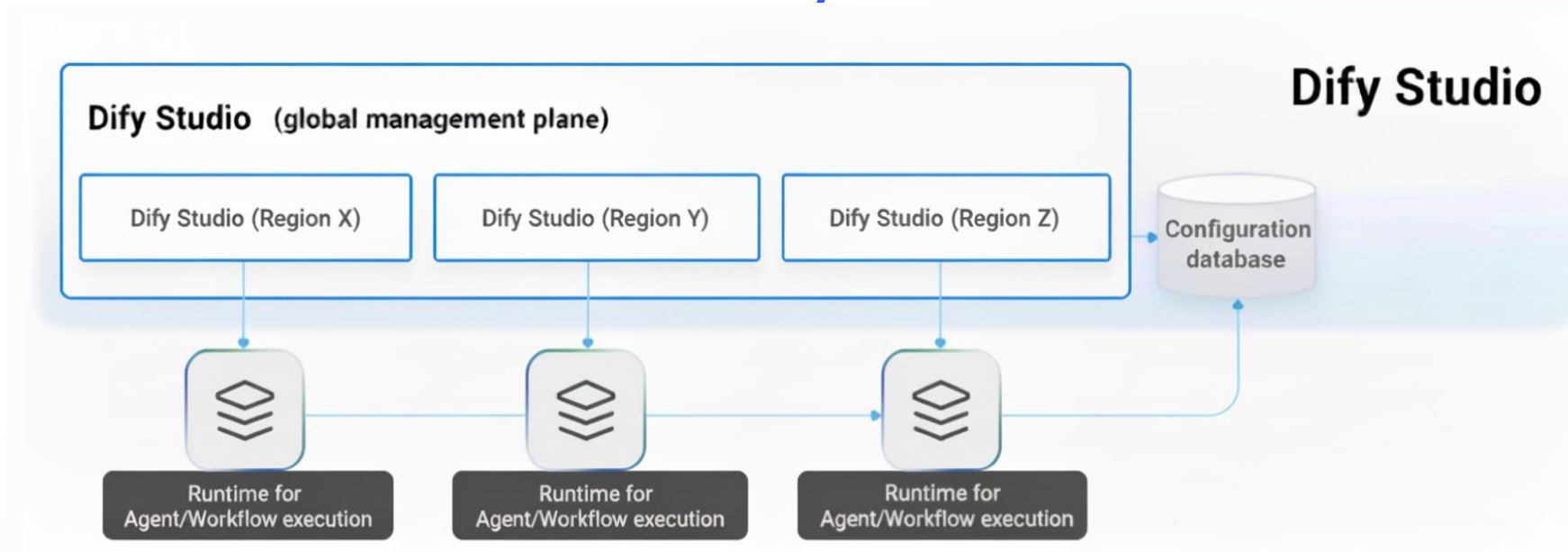
Architectural Blueprint



企业版多集群架构

- Centralized CI/CD & deployment control
- Unified endpoint for cross-region pages
- Shared in-memory cache for fast needs

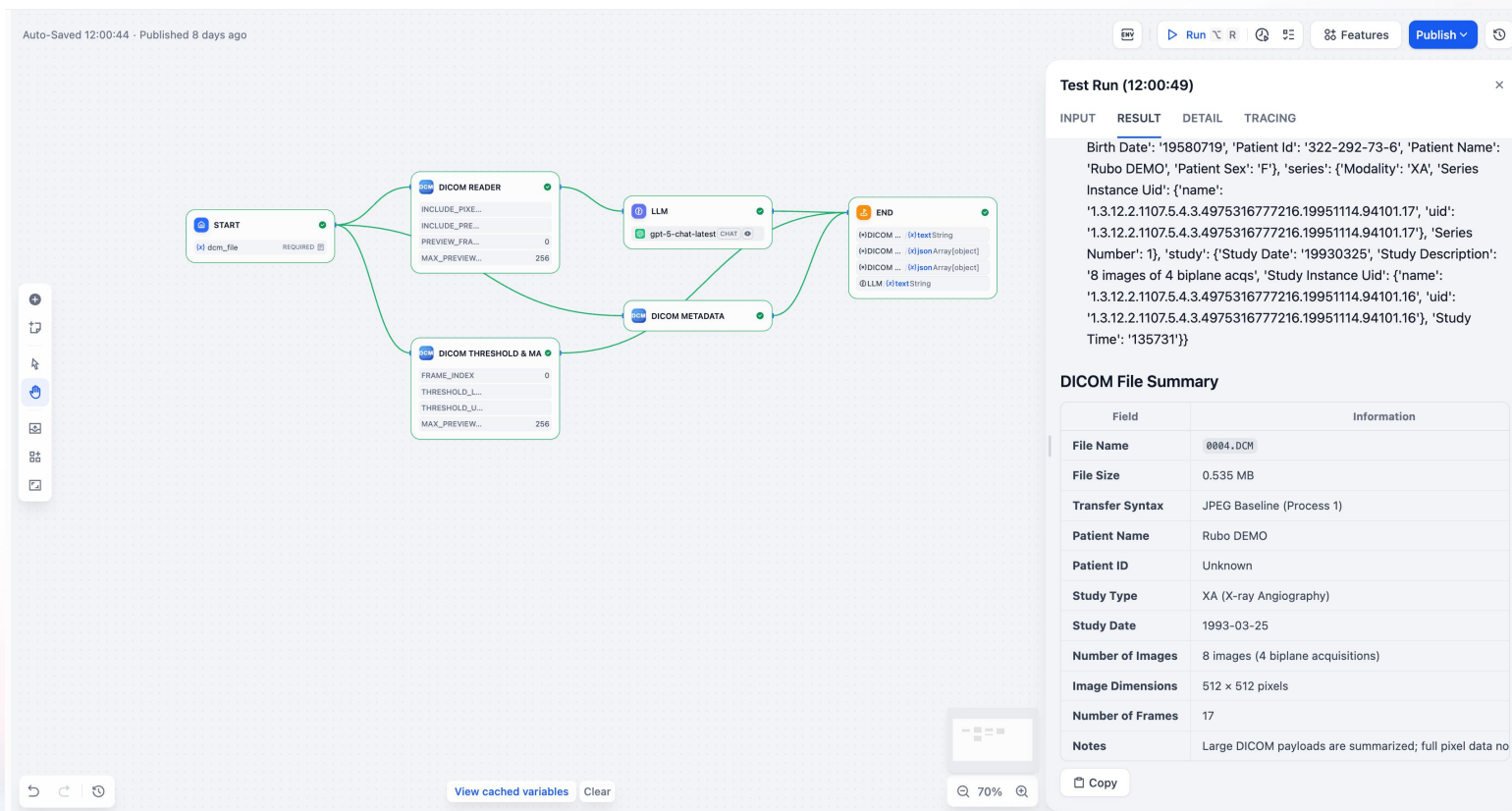
Runtime Layer in Action



Runtime layer 通过边缘计算原则，保证了工作流低延迟执行

- On-Demand Workflow Execution
- Independent Scaling
- Resilience and Version Control
- Robust Tenant Isolation

Demo DICOM Reader



编排 workflow (以下任一)

- Chatbot
- Agent
- Workflow
- Chatflow

Demo DICOM Reader

DCM Reader

Run Once

Run Batch

DICOM File

Local upload

Paste file link

0004.DCM

DCM • 547.36 KB

Clear

Execute

Workflow Process

START

DICOM READER

DICOM METADATA

DICOM THRESHOLD & MASK

END

RESULT

DETAIL

File 0004.DCM (0.535 MB, Transfer Syntax: JPEG Baseline (Process 1))

Patient: b'Rubo DEMO' (ID: unknown)

Study: XA on 19930325 — 8 images of 4 biplane acqs

Image size: 512 x 512 px, 17 frames

Large DICOM payloads are summarized so agents do not need to ingest the entire file.

- {'filename': '0004_mask_0.png', 'frame_index': 0, 'mask_pixels': 153284, 'mime_type': 'image/png'}
- {'file': {'filename': '0004.DCM', 'has_pixel_data': False, 'is_implicit_vr': False, 'is_little_endian': True, 'number_of_frames': 17, 'size_bytes': 560498, 'size_megabytes': 0.535, 'transfer_syntax_name': 'JPEG Baseline (Process 1)', 'transfer_syntax_uid': '1.2.840.10008.1.2.4.50'}, 'image': {'Bits Allocated': 8, 'Bits Stored': 8, 'Columns': 512, 'High Bit': 7, 'Image Type': ['DERIVED', 'PRIMARY', 'BIPLANE B'], 'Number Of Frames': 17, 'Photometric Interpretation': 'MONOCHROME2', 'Pixel Representation': 0, 'Rows': 512, 'Samples Per Pixel': 1, 'Sopinstance Uid': {'name': '1.3.12.2.1107.5.4.3.284980.19951129.170916.11', 'uid': '1.3.12.2.1107.5.4.3.284980.19951129.170916.11'}, 'patient': {'Patient Birth Date': '19580719', 'Patient Id': '322-292-73-6', 'Patient Name': 'Rubo DEMO', 'Patient Sex': 'F'}, 'series': {'Modality': 'XA', 'Series Instance Uid': {'name': '1.3.12.2.1107.5.4.3.4975316777216.19951114.94101.17', 'uid': '1.3.12.2.1107.5.4.3.4975316777216.19951114.94101.17'}, 'Series Number': 1}, 'study': {'Study Date': '19930325', 'Study Description': '8 images of 4 biplane acqs', 'Study Instance Uid': {'name': '1.3.12.2.1107.5.4.3.4975316777216.19951114.94101.16', 'uid': '1.3.12.2.1107.5.4.3.4975316777216.19951114.94101.16'}, 'Study Time': '135731'}}

POWERED BY Dify

运行 workflow

可以得到执行流程和结果
流程不会显示具体的细节
可以通过 WebApp 设置关闭

Demo DICOM Reader

Test Run (12:00:49)

INPUT RESULT DETAIL TRACING

>	 START	181.239 ms	
>	 DICOM READER	440.874 ms	
>	 DICOM METADATA	527.564 ms	
>	 DICOM THRESHOLD & MASK	1.049 s	
>	 LLM	295 tokens · 3.719 s	
>	 END	106.677 ms	

LLM

295 tokens · 3.719 s

INPUT

```
1 {
2   "#context#": "File `0004.DCM` (0.535 MB, Transfer Syntax:
   JPEG Baseline (Process 1))\nPatient: b'Rubo DEMO' (ID:
   unknown)\nStudy: XA on 19930325 - 8 images of 4 biplane
   acqs\nImage size: 512 x 512 px, 17 frames\nLarge DICOM
   payloads are summarized so agents do not need to ingest the
   entire file."
```

3 }

PROCESS DATA

1

```
3      "prompts": [
        do not need to ingest the entire file.\n\nOutput the
        context in the format of markdown.\n\n- Use table to
        display data\n- Use proper format to display
        information"
```

8

9

10

11

OUTPUT

1

```

**Transfer Syntax** | JPEG Baseline (Process 1) |\n|
**Patient Name** | Rubo DEMO |\n| **Patient ID** |
Unknown |\n| **Study Type** | XA (X-ray Angiography) |\n|
**Study Date** | 1993-03-25 |\n| **Number of Images** | 8
images (4 biplane acquisitions) |\n| **Image
Dimensions** | 512 x 512 pixels |\n| **Number of

```

Demo DICOM Reader

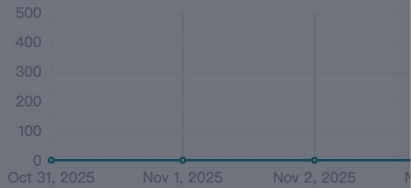
Monitoring

Last 7 Days

Total Messages

LAST 7 DAYS

0

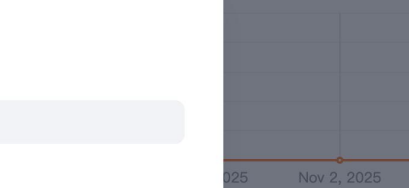


Oct 31, 2025 Nov 1, 2025 Nov 2, 2025

Active Users

LAST 7 DAYS

0

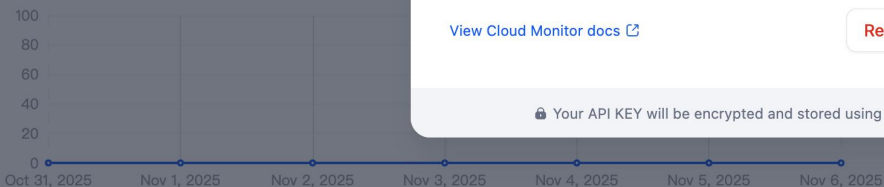


Oct 31, 2025 Nov 1, 2025 Nov 2, 2025

Token Usage

LAST 7 DAYS

0



Oct 31, 2025 Nov 1, 2025 Nov 2, 2025 Nov 3, 2025 Nov 4, 2025 Nov 5, 2025 Nov 6, 2025

IN SERVICE 云监控

Tracing

IN SERVICE

Capture the full context of app execution, including LLM calls, context, prompts, HTTP requests, and more, to a third-party tracing platform.

CONFIGURED

 云监控 IN USE View Config

The fully-managed and maintenance-free observability platform provided by Alibaba Cloud, enables out-of-the-box monitoring, tracing, and evaluation of Dify applications.

MORE PROVIDER

 arize Config

Enterprise-grade LLM observability, online & offline evaluation, monitoring, and experimentation—powered by OpenTelemetry. Purpose-built for LLM & agent-driven applications.

 DUNAMAI Config

Config Cloud Monitor

License Key *

bxst4ty*****01

Endpoint

http://tracing-analysis-dc-hz.aliyuncs.com

App Name

Cloud Testing

[View Cloud Monitor docs](#) Remove Cancel Save

🔒 Your API KEY will be encrypted and stored using PKCS1_OAEP technology.

Demo DICOM Reader

调用链分析

快捷筛选

状态

次数

☐ 正常

耗时

Ons

- 10ms

应用名称

次数

☐ Cloud Testing

Span 类型

次数

☐ TOOL
 ☐ TASK
 ☐ CHAIN
 ☐ LLM

接口名称

次数

☐ DICOM Metadata
 ☐ End
 ☐ DICOM Reader

INPUT

{ "dcm_file": { "dify_model_identity": "__dify_file__", "id": null, "tenant_id": "a06fddca-de98-465d-9d17-37f..." }

OUTPUT

{ "text": "File `0004.DCM` (0.535 MB, Transfer Syntax: JPEG Baseline (Process 1))\nPatient: b'Rubo DEMO' ... }

组件标签

gen_ai 6 llm 1

Trace详情

拓扑视图

全部

大模型

6	workflow	CHAIN	Cloud Testing	gen_ai	3.65 s
	Start	TASK	Cloud Testing	gen_ai	12.55 ms
	DICOM Reader	TOOL	Cloud Testing	gen_ai	333.07 ms
	DICOM Metadata	TOOL	Cloud Testing	gen_ai	441.44 ms
	DICOM Threshold & Mask	TOOL	Cloud Testing	gen_ai	1.08 s
	LLM	LLM	Cloud Testing	tokens: 294 llm	3.17 s
	End	TASK	Cloud Testing	gen_ai	72.04 ms

CHAIN workflow

应用名

Cloud Testing

接口名

workflow

IP

主机名

dify-tracing-ops-worker-f77767544-8jgh9

开始时间

2025-11-07 11:59:15.495

结束时间

2025-11-07 11:59:19.146

spanId

e7fb9dbb69bdb7b7

parentSpanId

0

状态码

正常

别名

附加信息

指标

日志

事件配置

Info

Attributes(12)

Resources(7)

Details(11)

Events(0)

Links(1)

input

6 items

"dcm_file": { 12 items

"dify_model_identity": string "__dify_file__"

"id": NULL

"tenant_id": string "a06fddca-de98-465d-9d17-37f53786dcb0"

"type": string "custom"

"transfer_method": string "local_file"

"remote_url": string "https://upload.dify.ai/files/d960337b-0569-4543-85dd-a9e8dac51965/file-prev-timestamp=1762487846&nonce=8909d2bb108b1e11dda73710cf6bc2f5&sign=vUuQqXjPdXoX9qlAC"

"related_id": string "d960337b-0569-4543-85dd-a9e8dac51965"

"filename": string "0004.DCM"

"extension": string ".dcm"

"mime_type": string "application/dicom"

"size": int 560498

"url": string "https://upload.dify.ai/files/d960337b-0569-4543-85dd-a9e8dac51965/file-prev-timestamp=1762487955&nonce=eb44d5aed36f3a45cc154e30f5afba16&sign=BeU5ZK5KrCpdflo4RoIU3KM%3D"

output



施子腾

阿里云可观测工程师

Agent & AI Infra & Observability

可观测性——两个视角

Agentic 应用开发者

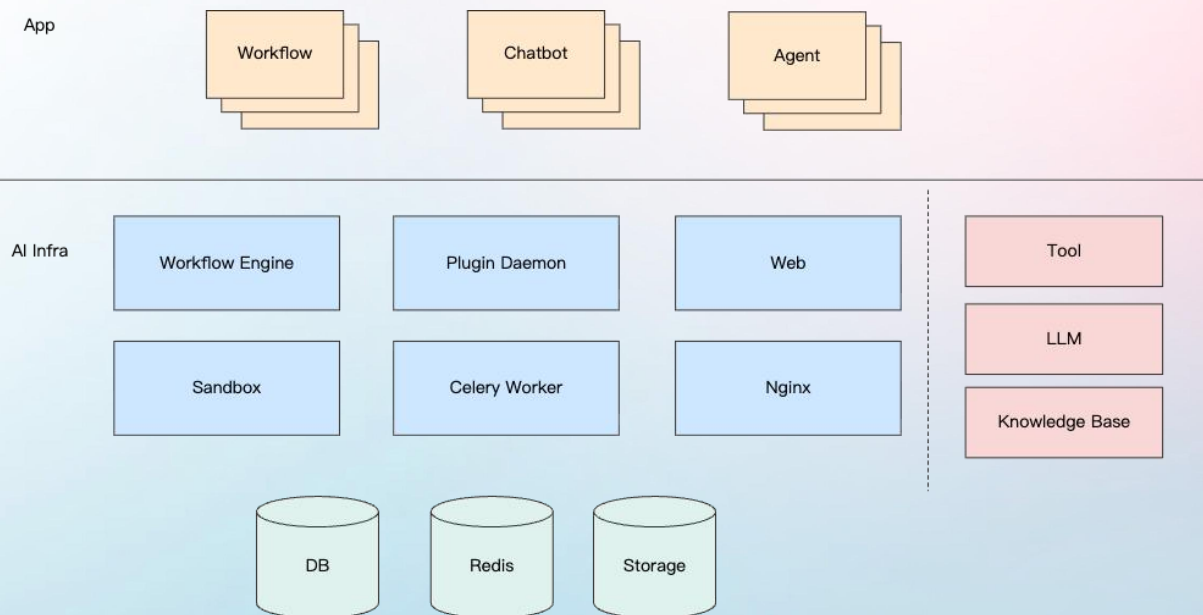
观测对象：Chatbot/Workflow 整体 & RAG、Tool、LLM 等各步骤的执行

关注点：效果、性能、多轮会话

Dify 平台运维方

观测对象：Dify 系统组件、集群基础设施、上下游依赖

关注点：全链路追踪、性能&负载



Dify 可观测——三种类型

Dify 内置

特性：Dify内置，基于引擎执行明细，DB存储

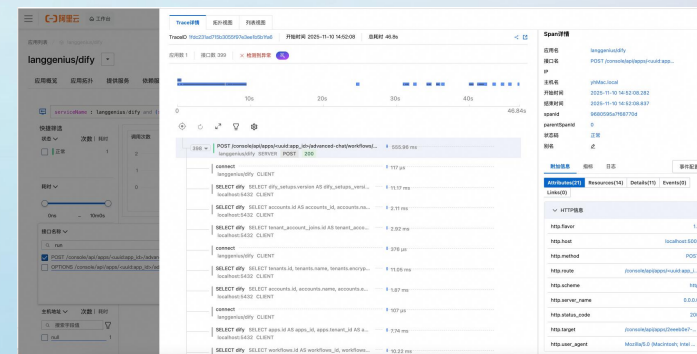
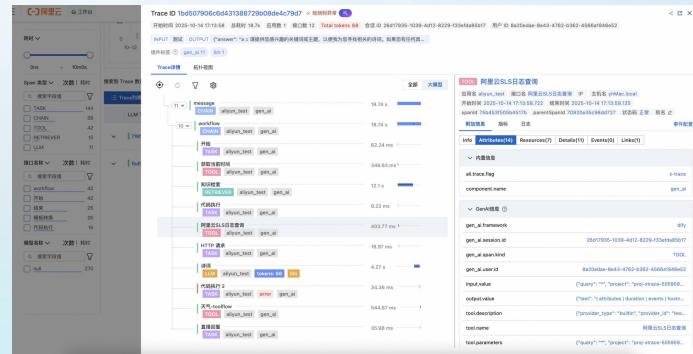
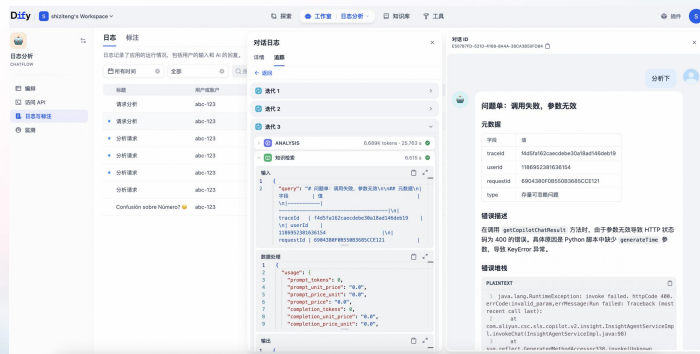
- 优势：开发调试使用便捷
- 痛点：分析能力和性能

Dify 集成

- 特性：事件后聚合机制，开放第三方对接
- 优势：第三方存储、分析
- 痛点：数据采集粒度受限、缺少全链路追踪能力

全链路 Trace

- 方案：Sentry & OTel
- 面向 Dify 集群各组件及上下游依赖的全链路可观测
- 痛点：采集数据有限，无法串联上下游



全景可观测——挑战

迭代迅速，代码变动频繁

Dify社区贡献者 1000+，发版周期 一两周至数天

Dify实现频繁变动，纯无侵入探针方式难以跟上Dify发版节奏

组件众多，执行链路复杂

请求经过网关入口、执行引擎、插件引擎、代码沙箱、插件运行时等多个组件，关联 Celery 后台任务队列，插件运行时受插件引擎的私有协议管理，链路复杂

两套机制，实现方式割裂

模型应用层监控使用Dify自定义的异步后聚合方式，和Infra层监控的 OTeI 生态差异大

LoongSuite AI采集套件



源于开源

社区贡献与回馈

LoongSuite 可观测采集套件 **LoongSuite**

LoongSuite Python Agent

LoongSuite Go Agent

LoongSuite Java Agent

LoongCollector

OTLP

OTLP

OTLP

OTLP

OTLP

开源存储

开源控制台

云服务

云服务控制台

性能分析

成本分析

质量分析
(评估)

AI Agent

基于 LoongSuite 的高质量数据采集

拥抱开源，面向大模型应用量身打造



一键安装接入

K8S环境支持修改YAML实现简单自动接入，initContainer机制注入探针，支持批量规模化自动接入



埋点更精细

针对框架埋点丰富优化，提供丰富的指标，支持用户添加自定义属性信息并在整条trace中透传



性能稳定性

埋点上报开销优化，降低应用性能影响，通过管控策略可以进一步降低开销在可控范围

LoongSuite Python 无侵入探针

Monkey Patch

1. 原始函数定义

```
def hello(self):  
    print("Hello, World!")
```

2. 包装函数定义

```
def wrapped_hello():  
    // do something before  
    print("Hello, before")  
  
    hello()  
  
    // do something after  
    print("Hello, after")
```

3. 完成函数的替换

```
hello = wrapped_hello
```

Python 探针

4. 最终结果

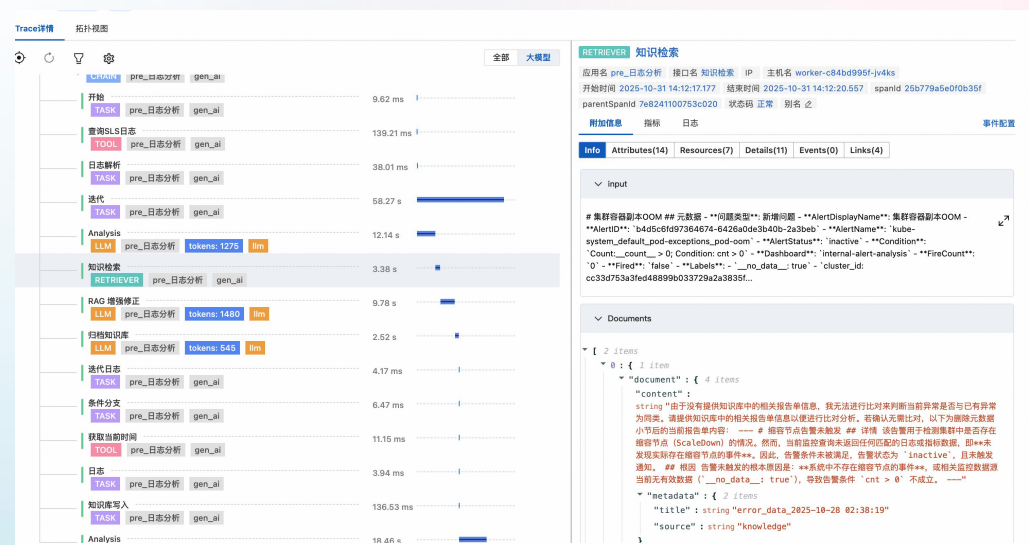
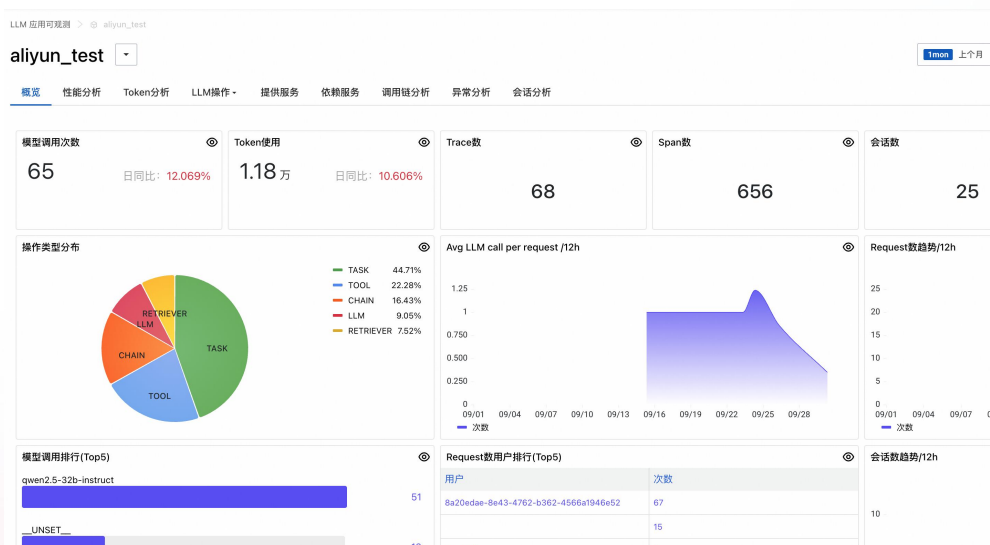
```
hello()  —————> wrapped_hello()  
Hello, World!    Hello, before  
                  Hello, World!  
                  Hello, after
```

Dify 全景路可观测方案

三方集成+无侵入探针

Dify 组件	可观测方案
Workflow 应用	Dify 官方集成的 云监控 应用追踪
执行引擎 Api	Python 探针
插件引擎 Plugin-Daemon	Go 探针
(可选) 任务队列 Worker	OTel 接入
(可选) 代码沙箱 Sandbox	Go 探针
(可选) 入口网关 Nginx	OTel 接入
(自动) 插件运行时 Plugin	Python 探针, 由 Plugin-Daemon 探针自动拉起

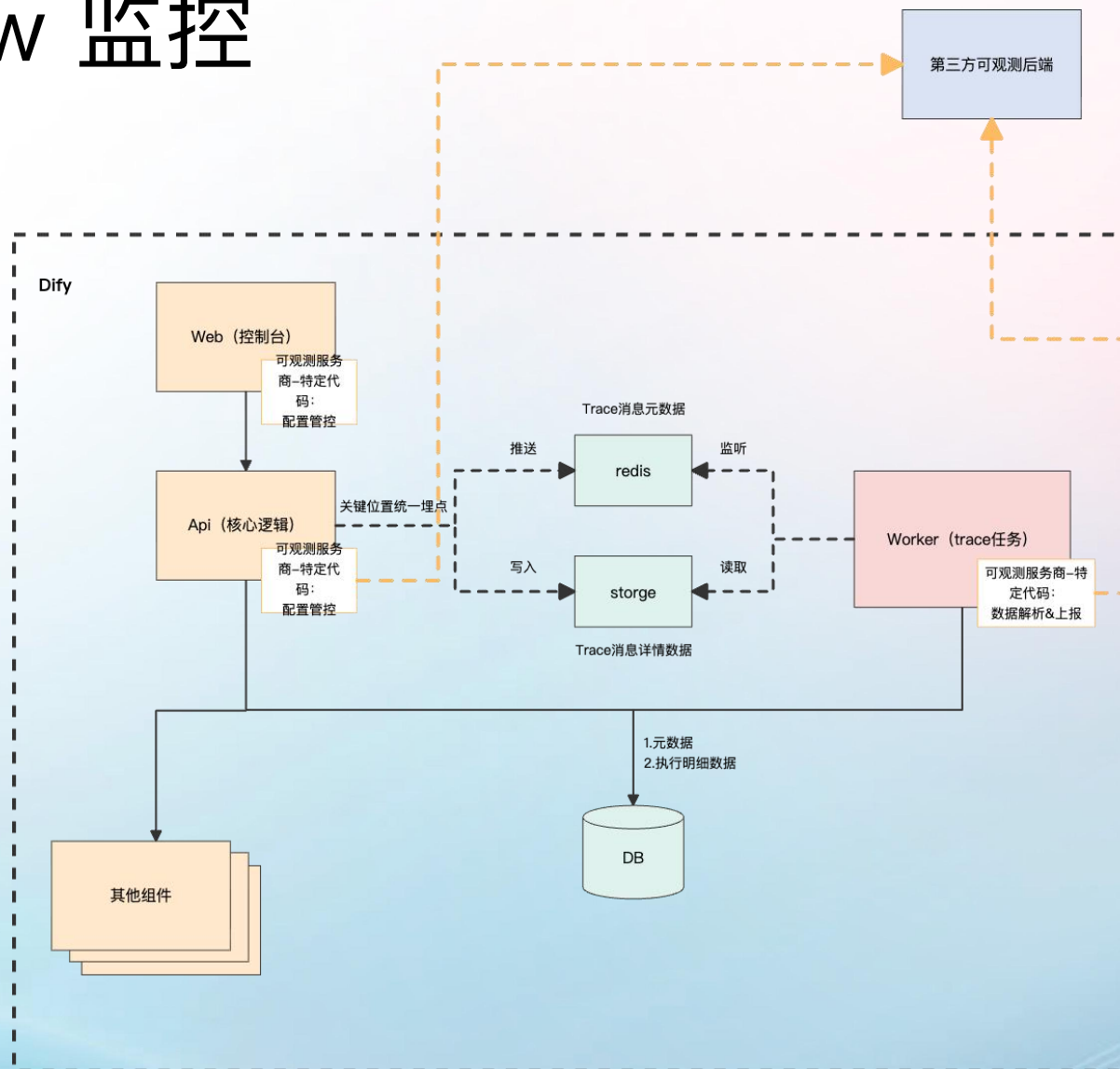
监控 Workflow 应用



集成 Workflow 监控

Dify 第三方监控集成

- 后聚合 Trace 数据
- 执行日志 转换 Trace
- OTel 格式上报
- Context 信息透传

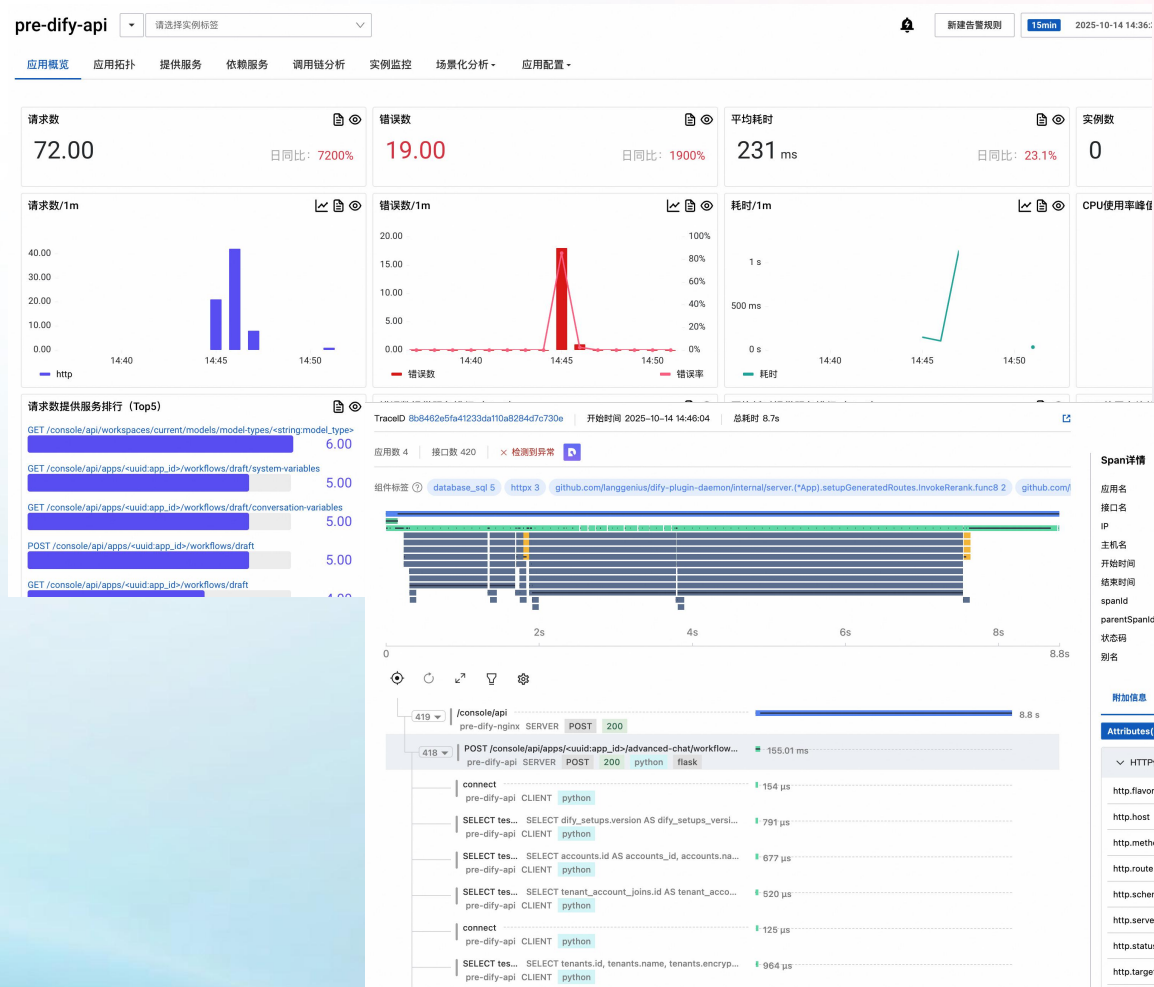


监控执行引擎

Dify-api 是核心执行引擎，负责 Agent/Workflow 执行调度

Python 无侵入探针

- 框架层埋点
 - HTTP/Flask/Redis/DB
- Dify增强埋点
 - RAG/Workflow/LLM
- 透传Context



监控插件引擎

Dify-Plugin-Daemon 是插件引擎，Dify 的 LLM、插件、Agent 策略的执行都依赖 Plugin-Daemon

Go 无侵入探针

- 编译时插桩

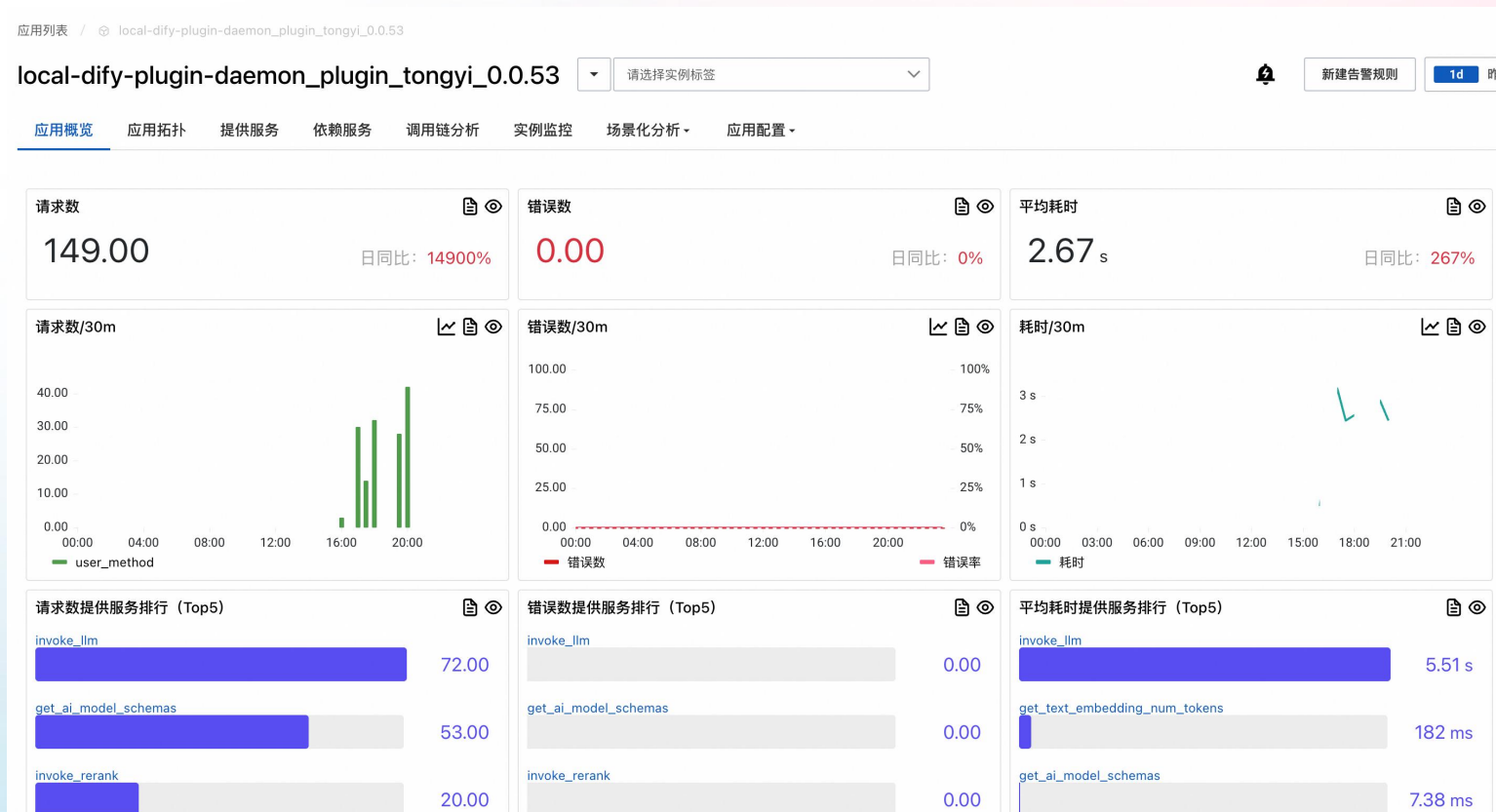


监控插件运行时

LLM、工具、Agent执行策略以插件方式运行
插件引擎以子进程方式拉起插件运行时

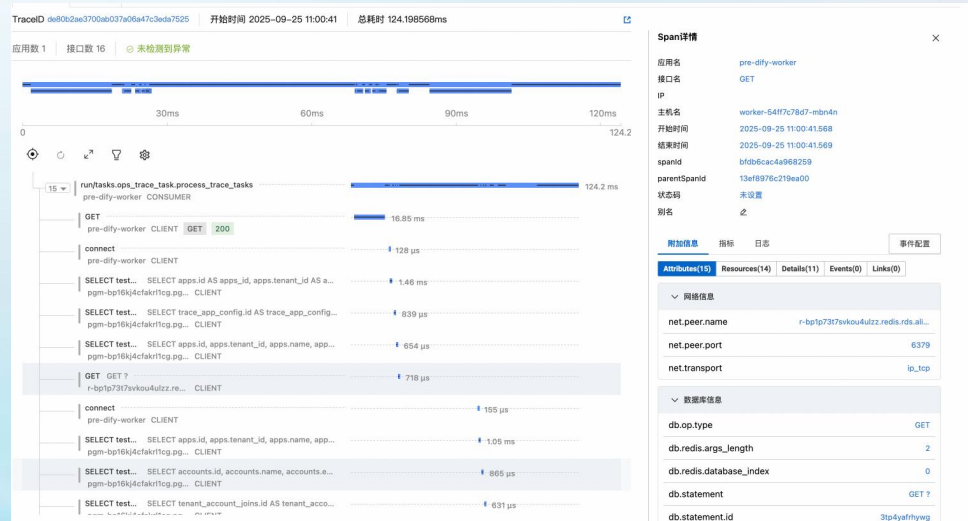
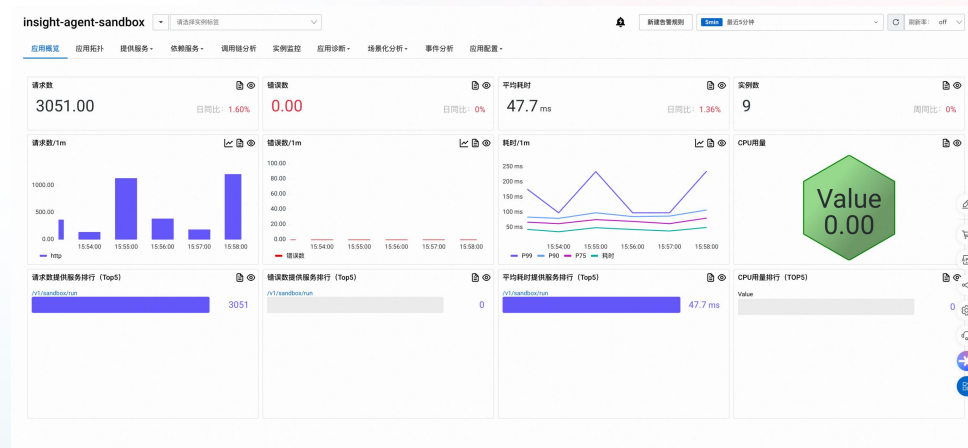
Python 无侵入探针

- 由插件引擎上的Go探针自动拉起



监控辅助组件

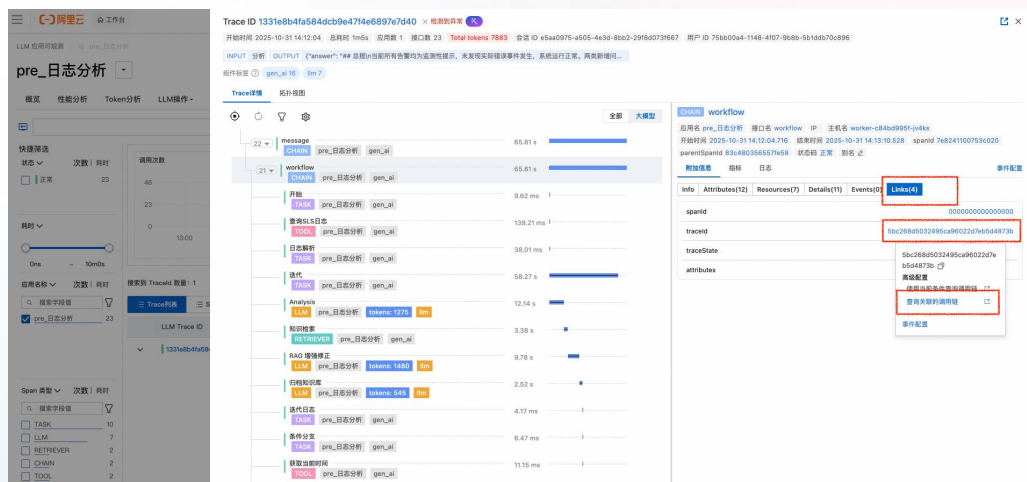
- 代码沙箱 Sandbox — Gin框架
 - Go 无侵入探针
- 任务队列 - Celery 任务
 - OTel 无侵入探针
- 网关 - Nginx
 - OTel 无侵入探针



关联 LLM Trace 和 Infra Trace

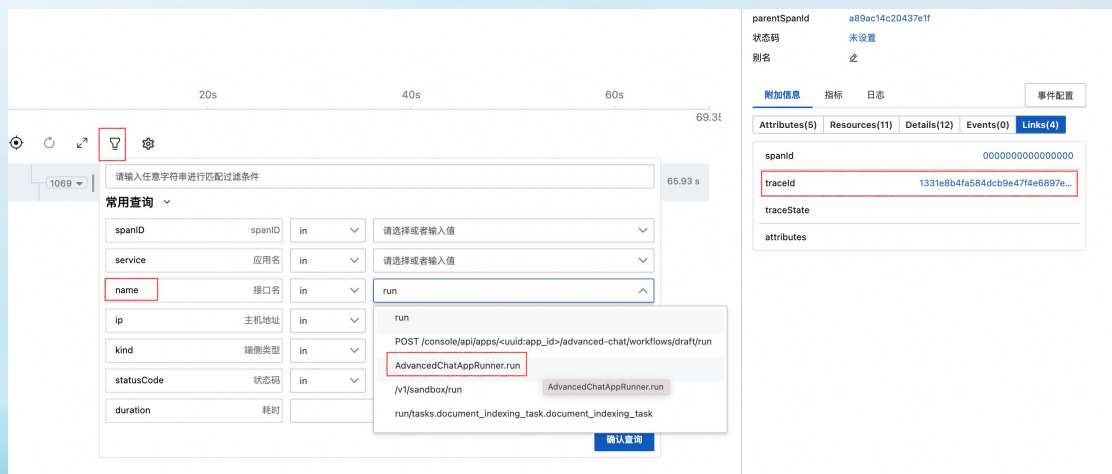
从 LLM Trace 跳转 微服务 Trace

- 每个Span的Trace Link 记录了关联的Infra层Trace



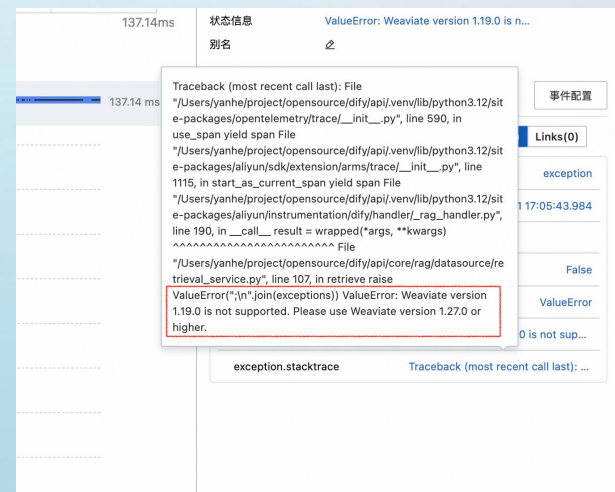
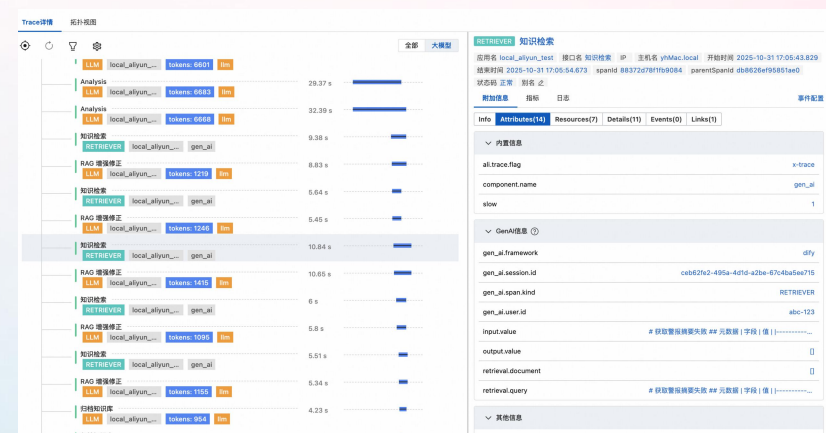
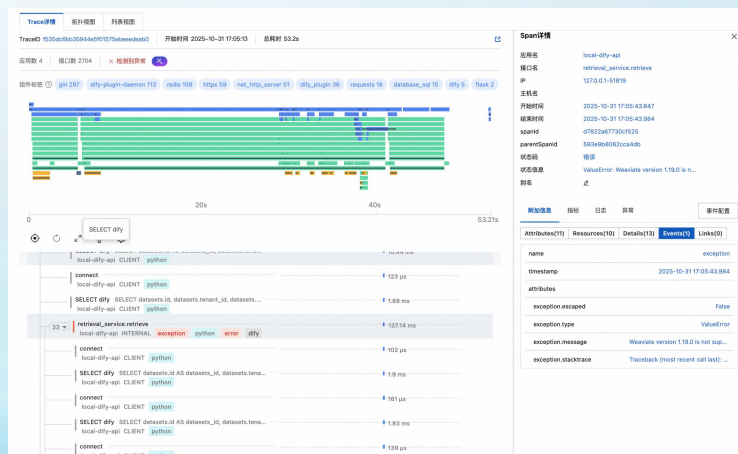
从 Infra Trace 跳转 LLM Trace

- AppRunner Span Link了关联的LLM层Trace



workflow 层面的异常通常在 LLM Trace 或 Dify 日志追踪里可以看到直观的报错信息，但还有一类错误是底层 Dify 引擎的配置不当或内部 bug 引起的。

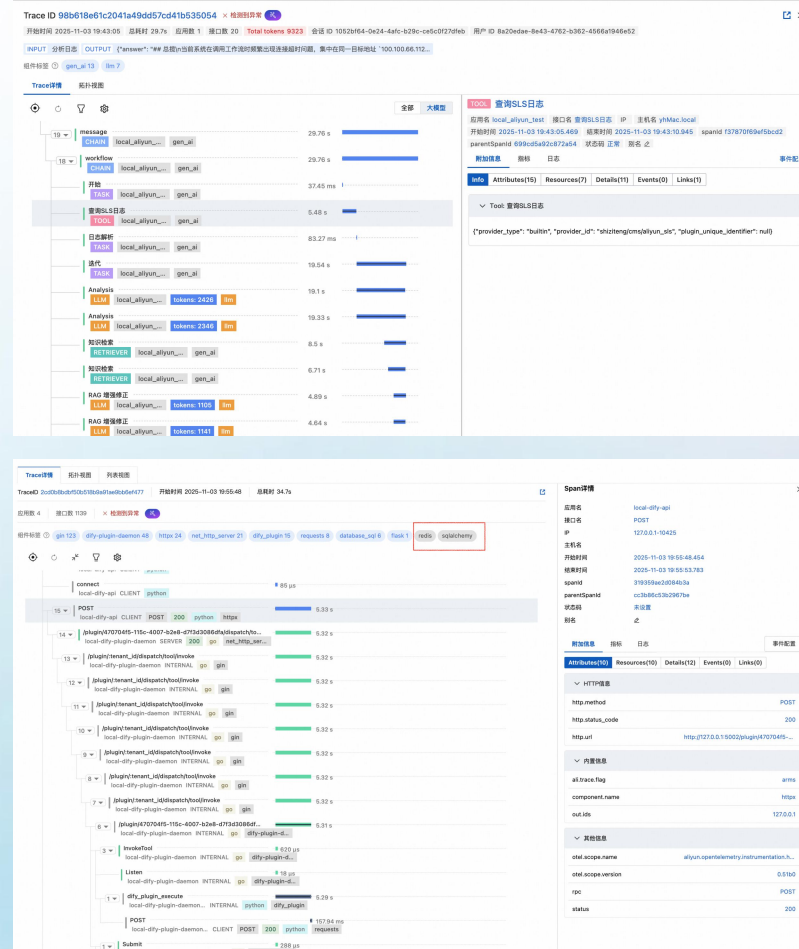
无侵入探针提供了对这类错误的定位分析能力。



实践——定位插件慢调用

管理插件的 Plugin-Daemon 和插件运行时是难以定位故障的黑盒。一次 Workflow 调用链路涉及 Nginx、Api、Plugin-Daemon、Plugin 运行时和模型网关等多个组件。

云监控提供的全链路追踪能力可以串联上下游的完整链路，定位插件内的错慢异常。



Dify全链路可观测最佳实践



一站式解决 从LLM应用调优评估到全链路追踪

Agentic 应用开发者

使用 Dify监控集成 从Agentic应用维度追踪
N个应用

Dify 平台运维方

使用 无侵入探针 从Infra层追踪 1套Dify基
础设施

迭代迅速，代码变动频繁？

✓ 使用Dify监控集成方式上报易变的Workflow层埋点

组件众多，执行链路复杂？

✓ Python、Go、Java 多语言探针支持

✓ 插件运行时探针 自动拉起/透传

两套机制，实现方式割裂？

✓ Trace Linke 关联 LLM 应用层Trace 和 Infra层 Trace

更细致的埋点，更原生的体验？

➡ 贡献Workflow引擎埋点到Dify仓库